Assignment – 4 (230107070)

Given Data:

Feed conditions:

- Pressure: 3 bar
- Vapor fraction: 0.5
- Composition: equimolar mixture of C2H5OH and C6H6
- Flow rate: 100 kmol/hr

Column 1: Operating pressure 3 bar

Column 2: Operating pressure 1 bar

Recycle stream: Compressed from 1 bar to 3 bar before entering Column 1

Condenser type: Partial vapor condenser

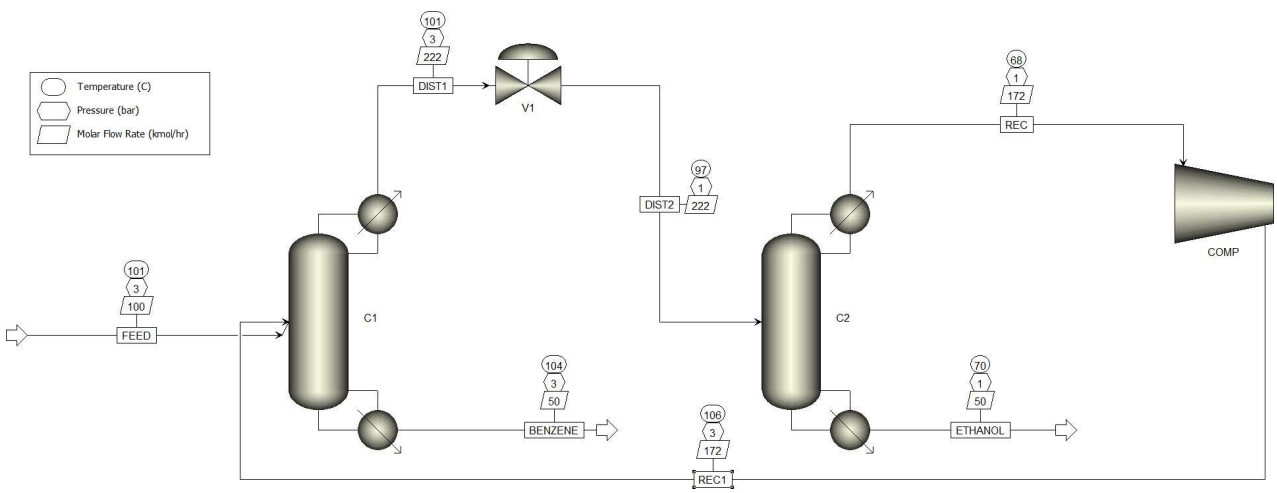


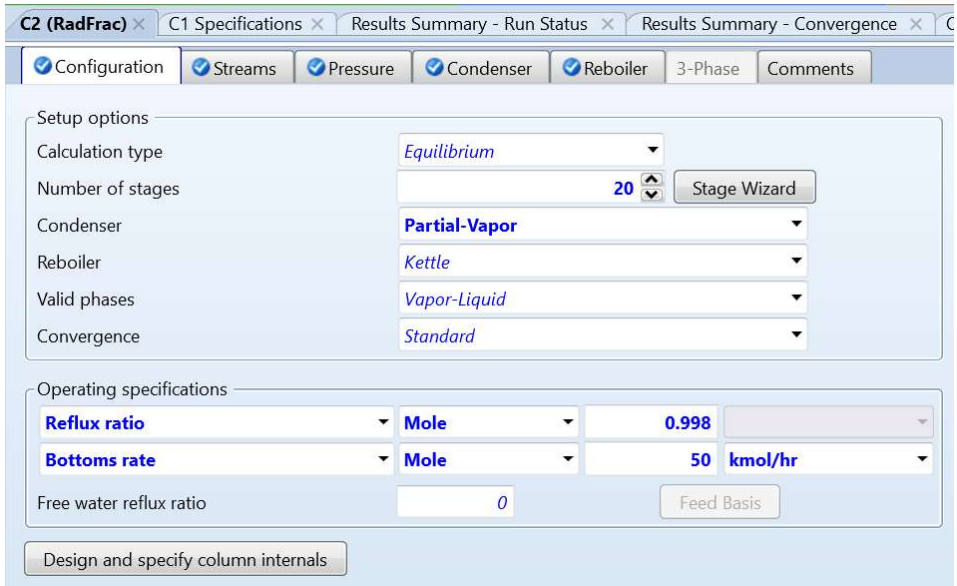
Diagram of flowsheet

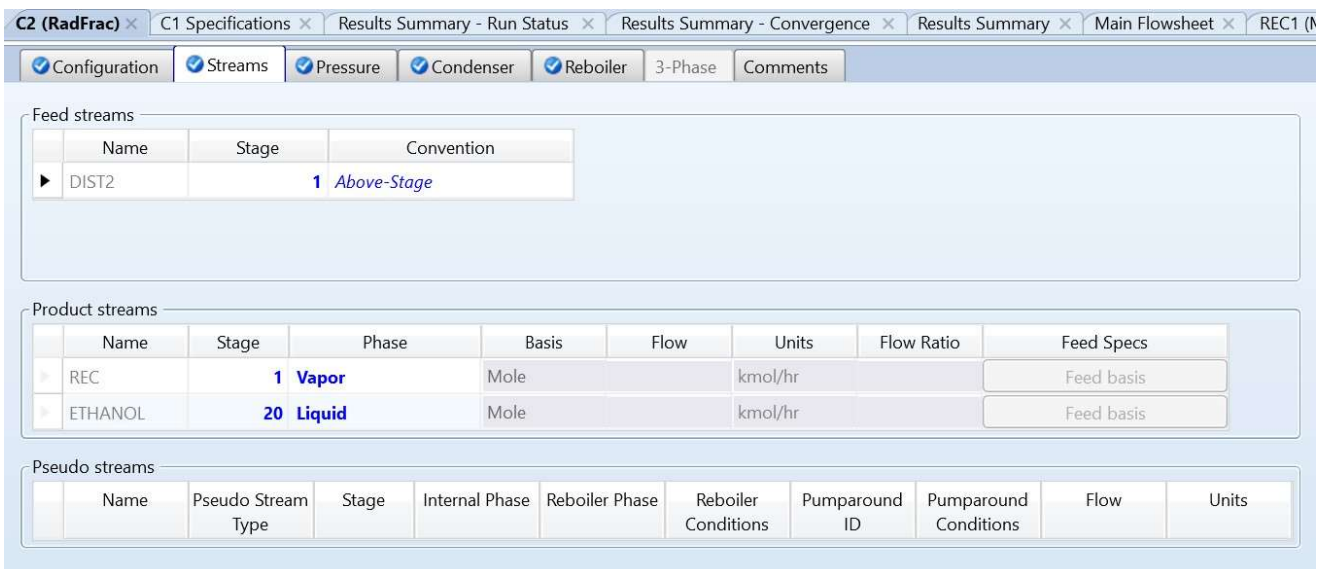
Column-2 (C2)

Number of stages = 20

Reflux ratio = 0.998

Feed stage location = 1



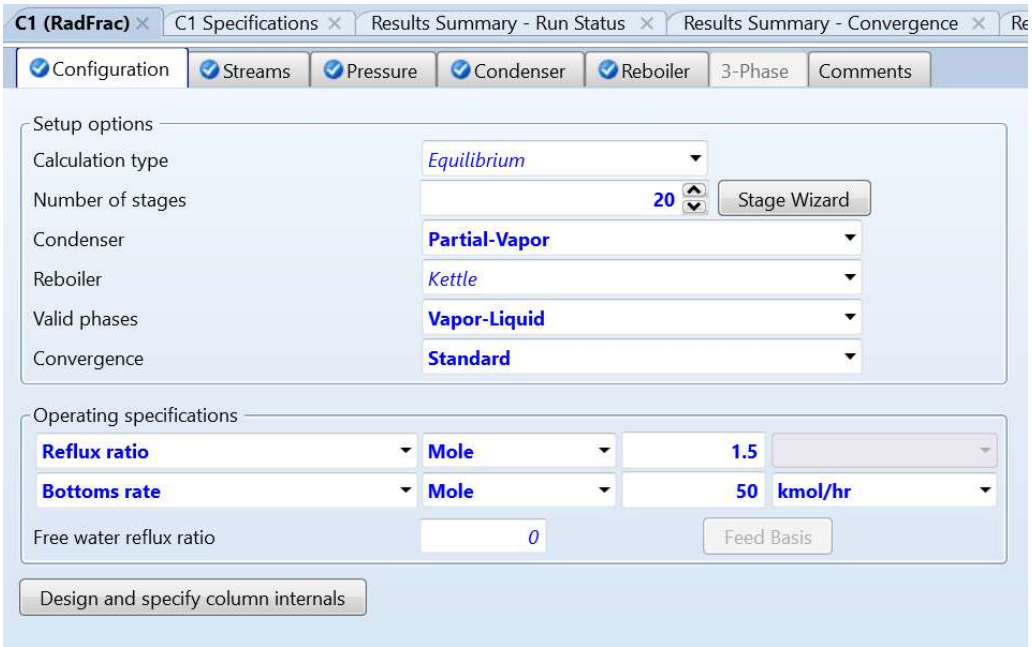


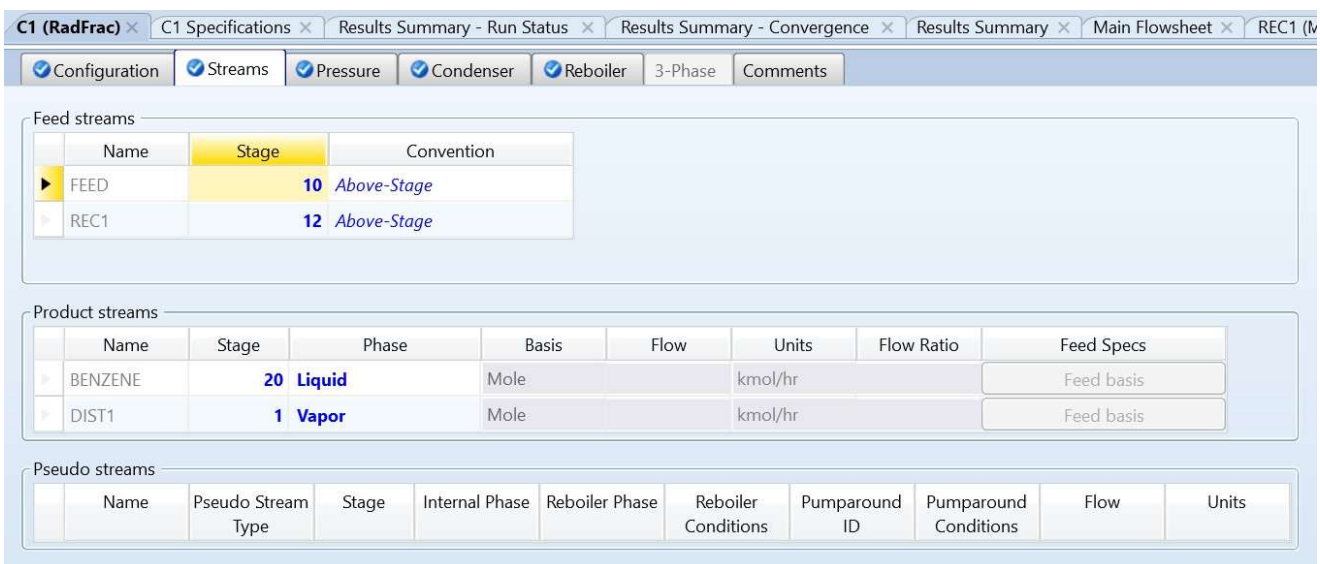
Column- (C1)

Number of stages = 20

Reflux ratio = 1.5

Feed stage location = 10





Result Stream

Composition of recycle stream (mol frac)

C6H6 => 0.516163

C2H5OH => 0.483837

Mole flow rate of recycle stream => 172.163 kmol/hr

REC1 (MATERIAL) - Results (Default) | C1 (RadFrac) | C1 Specifications | Results Summary - Run Statu

Material | Vol.% Curves | Wt. % Curves | Petroleum | Polymers | Solids | Status

	Units	REC1
Molar Enthalpy	cal/mol	-15411.2
Mass Enthalpy	cal/gm	-246.149
Molar Entropy	cal/mol-K	-41.4799
Mass Entropy	cal/gm-K	-0.66252
Molar Density	mol/cc	0.000100314
Mass Density	gm/cc	0.00628058
Enthalpy Flow	cal/sec	-737011
Average MW		62.6093
+ Mole Flows	**kmol/hr**	**172.163**
− Mole Fractions		
C2H5OH		0.483837
C6H6		0.516163
+ Mass Flows	**kg/hr**	**10779**
+ Mass Fractions		
Volume Flow	l/min	28604.1
+ Vapor Phase		
<add properties>		

Results Summary - Streams (All) | REC1 (MATERIAL) - Results (Default) | C1 (RadFrac) | C1 Specifications | Results Summary - Run Status | Results Summary - Convergence | Main Flowsheet | REC1 (

Material | Heat | Load | Work | Power | Vol.% Curves | Wt. % Curves | Petroleum | Polymers | Solids

	Units	BENZENE	DIST1	DIST2	ETHANOL	FEED	REC	REC1
Mass Enthalpy	cal/gm	2.07964	-351.482	-351.482	-961.713	-330.173	-258.239	-246.149
Molar Entropy	cal/mol-K	-55.0634	-43.1258	-41.0138	-73.3718	-52.0655	-41.4804	-41.4799
Mass Entropy	cal/gm-K	-0.763839	-0.715865	-0.680807	-1.40848	-0.838531	-0.66252	-0.66252
Molar Density	mol/cc	0.0108393	0.00010189	3.30388e-05	0.0148293	0.000202169	3.61199e-05	0.000100314
Mass Density	gm/cc	0.781381	0.00613814	0.00199035	0.772501	0.0125529	0.00226146	0.00628058
Enthalpy Flow	cal/sec	2082.18	-1.30671e+06	-1.30671e+06	-695810	-569470	-773219	-737011
Average MW		72.0877	60.2429	60.2429	52.0928	62.0913	62.6099	62.6093
+ Mole Flows	**kmol/hr**	**50**	**222.163**	**222.163**	**50**	**100**	**172.163**	**172.163**
− Mole Fractions								
C2H5OH		0.18805	0.557682	0.557682	0.812019	0.5	0.483817	0.483837
C6H6		0.81195	0.442318	0.442318	0.187981	0.5	0.516183	0.516163
+ Mass Flows	**kg/hr**	**3604.38**	**13383.8**	**13383.8**	**2604.64**	**6209.13**	**10779.1**	**10779**
+ Mass Fractions								
Volume Flow	l/min	76.8806	36340.4	112072	56.1949	8243.94	79440.6	28604.1
+ Vapor Phase								
+ Liquid Phase								
<add properties>								

The solution is for 0.812019 mol frac of C2H5OH

DETERMINE THE FOLLOWING:

a) What is the steady state reflux ratio required?

The steady-state reflux ratio required for the system is **1.5 for C1** and **0.GG8 for C2**.

b) What is the number of stages required?

Stage calculations were performed based on equilibrium data.

The number of stages:

Column C1: **20**

Column C2: **20**

c) Locate the feed stage position in both the columns.

The feed stage position:

Column C1 Feed Stage: **10**

Column C2 Feed Stage: **1**

d) Recycle stream flow rate and composition?

Composition of recycle stream (mol frac)

C6H6 => 0.516163

C2H5OH => 0.483837

Mole flow rate of recycle stream => 172.163 kmol/hr